

Lab 3 – Written Justification

Faculty Research Grant & Publication Tracker

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Course: Software Engineering	Lab: 3 – Component Modelling & Architectural Pattern Selection

Problem Statement #09 – Campus & Academic Operations

Architecture Selection

We chose Layered Architecture for the Faculty Research Grant & Publication Tracker because it provides a clear separation between the user interface, business logic, and data management. The architecture consists of Presentation, Business, and Data layers, making the system easier to understand, maintain, and manage.

Architectural Choice

Layered Architecture is appropriate for this system because the application contains distinct responsibilities such as managing research publications and citation metrics, tracking grant allocations and expenses, handling co-author approval workflows, and storing persistent research and financial data.

Reason 1 – Separation of Responsibilities

The Presentation Layer provides access through the Web Portal, while the Business Layer handles application logic, research management, grant management, and notifications. The Data Layer manages persistent information. This separation makes individual components easier to understand, modify, and maintain.

Reason 2 – Management of Research and Financial Workflows

The system needs to manage grant fund allocations, equipment procurement expenses, publication metrics, co-author approval workflows, and fund burn-up analytics. Separating these functions into business components allows each responsibility to be handled independently while the Application Service coordinates the overall workflow.

Security Advantage

The layered structure allows security controls to be applied at appropriate layers. Sensitive financial approvals and journal status modifications can be protected through the application and data layers, while audit information can be maintained in the database. This supports the requirement for an immutable, tamper-evident audit ledger.

Performance Benefit

Separating business processing from data management allows database access through a dedicated Data Access Interface and keeps business processing within the Business Layer. This organized separation can reduce unnecessary interaction between components and makes individual parts easier to optimize.

Conclusion

Therefore, Layered Architecture is selected because it provides clear separation of concerns, supports the system's research and financial workflows, provides a suitable structure for security controls, and offers an organized approach to performance and maintenance.